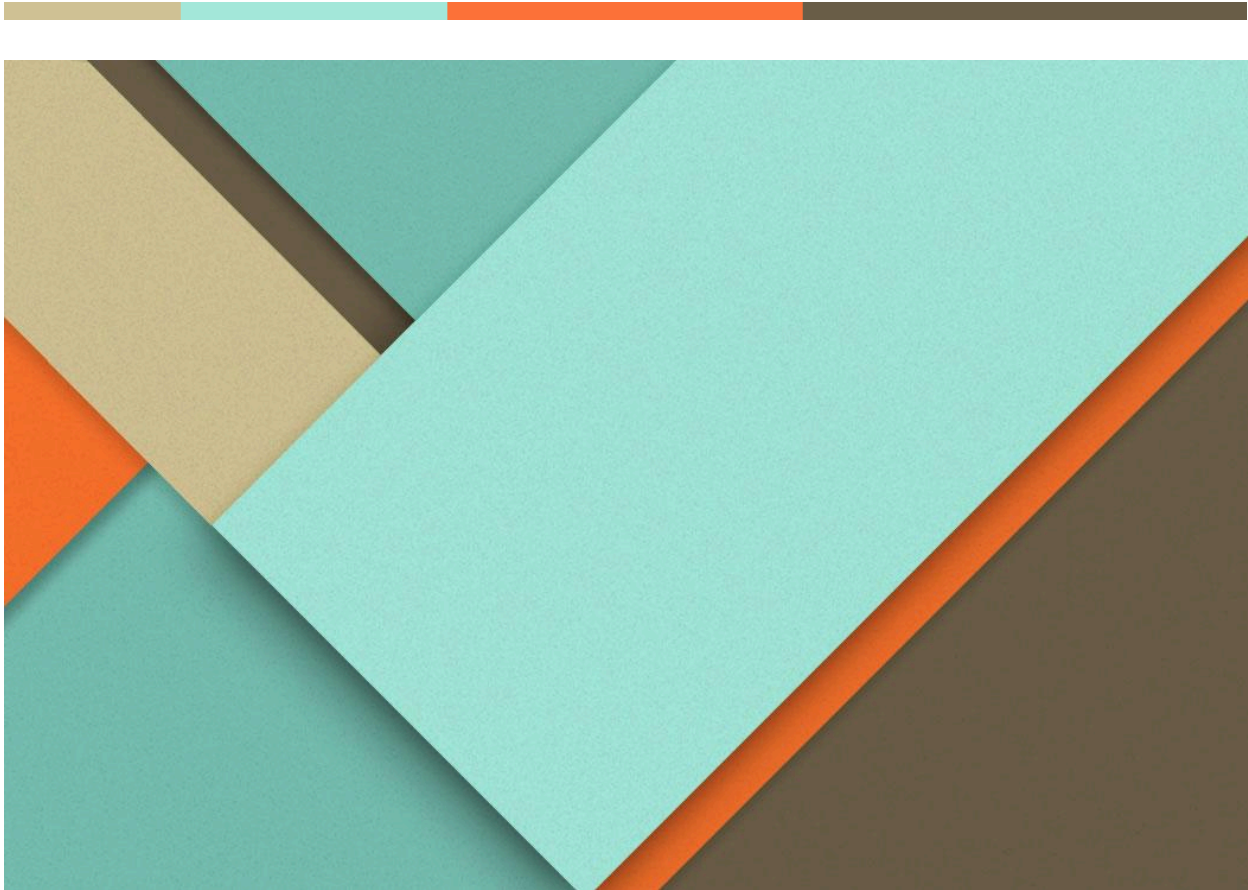




PIN FINAL

11.08.2025

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Resumen:

El objetivo de este PIN, será utilizar github actions para desplegar un sitio estático en un cluster de AWS EKS con AWS ECR como repositorio de imágenes y con Terraform como proveedor de IaC. Se agregan herramientas de monitoreo (Prometheus + Grafana) y se crea una máquina virtual EC2 para realizar tareas de mantenimiento más flexible sobre el cluster EKS.

Repositorio:

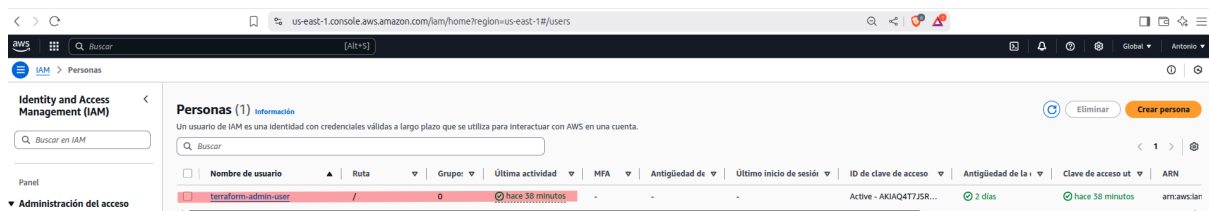
<https://github.com/alefinir/PIN-mundose.git>

Evidencia:

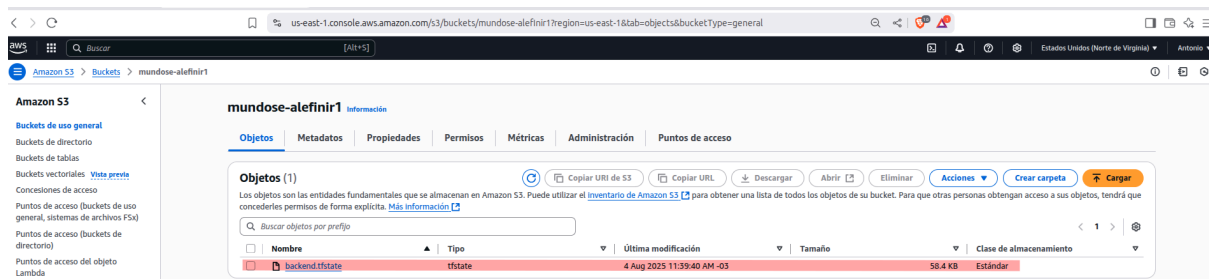
Recursos para el backend de Terraform:

Para que el backend de terraform funcione se va a crear un usuario de acceso AWS CLI (sin consola), también un bucket que guarde el archivo de estado de Terraform y una base de datos para bloquear el acceso a otros usuarios mientras terraform este corriendo (terraform plan / terraform apply).

- Usuario:



- Bucket:



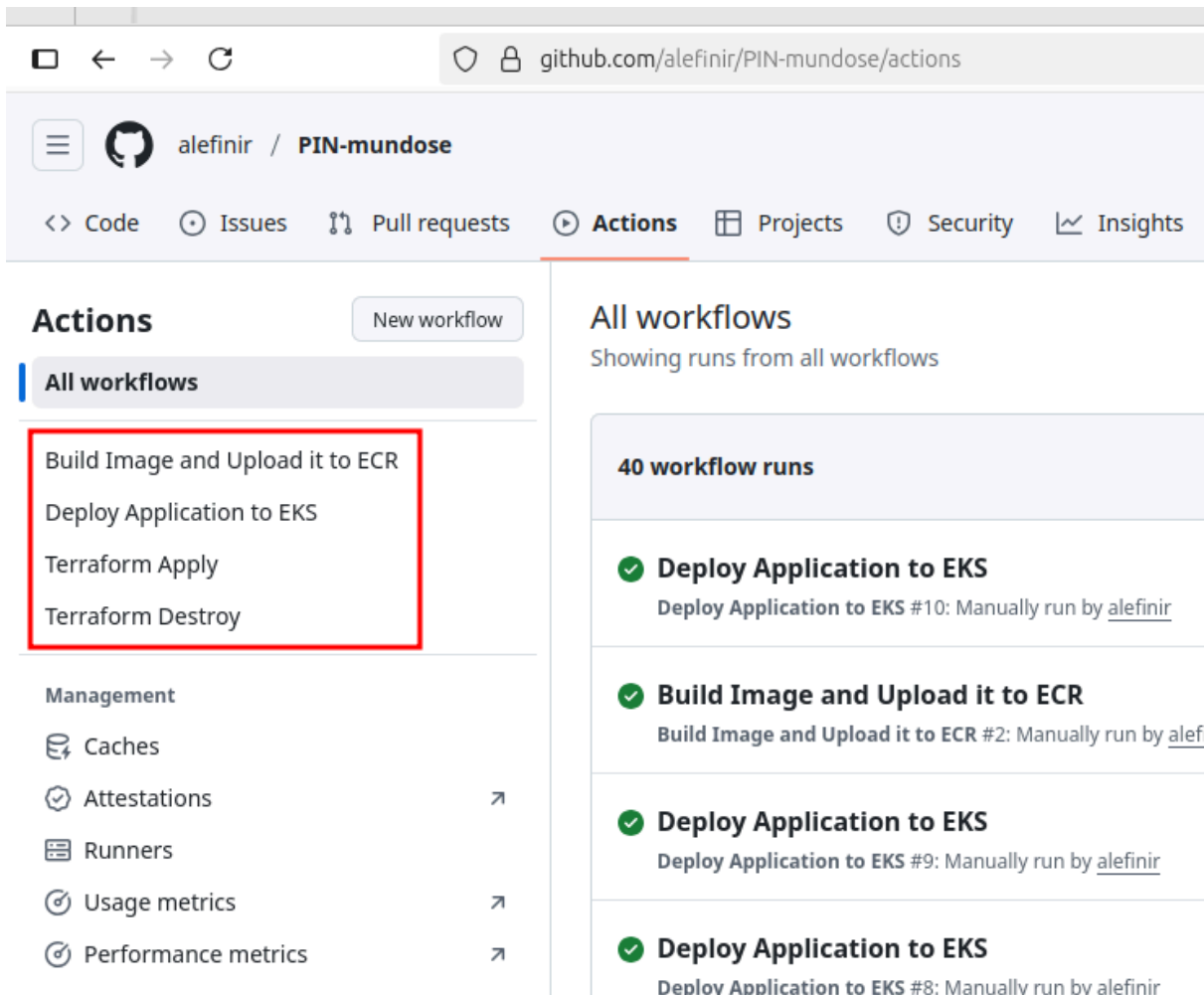
- Tabla de Dynamodb:



GitHub Actions - Workflows:

Hemos creado 4 Workflows para este proyecto, normalmente se ejecutan en este orden:

1. Creación del Cluster EKS
2. Creación de la Imagen y subida al repositorio ECR
3. Deploy de la imagen al Cluster y actualización del deployment/servicio
4. Borrado de todo el cluster e infraestructura relacionada.



Link al repo:

<https://github.com/alefinir/PIN-mundose/tree/main/.github/workflows>

Terraform Apply:

Este workflow nos permite deployar la infraestructura escrita en archivos de terraform (nuestro proveedor de IaC), primero se formatea el archivo (comando terraform fmt), luego se presentan los cambios (terraform plan) y por último se escriben los cambios en la nube de amazon (se deploya un cluster de EKS):

The screenshot shows a GitHub Actions workflow run for the job 'terraform.apply'. The workflow is titled 'terraform.apply' and shows it succeeded 3 hours ago in 12m 5s. The job output displays the Terraform apply process, showing the creation of various AWS EKS resources. The output includes a list of resources being created, such as 'aws_eks_node_group.eks_node_group', and a final summary indicating that the cluster was successfully created. The output also shows the cluster endpoint and name.

```

terraform.apply
succeeded 3 hours ago in 12m 5s

Terraform apply
937 aws_eks_node_group.eks_node_group: Still creating... [00m40s elapsed]
938 aws_eks_node_group.eks_node_group: Still creating... [00m50s elapsed]
939 aws_eks_node_group.eks_node_group: Still creating... [01m00s elapsed]
940 aws_eks_node_group.eks_node_group: Still creating... [01m10s elapsed]
941 aws_eks_node_group.eks_node_group: Still creating... [01m20s elapsed]
942 aws_eks_node_group.eks_node_group: Still creating... [01m30s elapsed]
943 aws_eks_node_group.eks_node_group: Still creating... [01m40s elapsed]
944 aws_eks_node_group.eks_node_group: Still creating... [01m50s elapsed]
945 aws_eks_node_group.eks_node_group: Still creating... [02m00s elapsed]
946 aws_eks_node_group.eks_node_group: Still creating... [02m10s elapsed]
947 aws_eks_node_group.eks_node_group: Still creating... [02m20s elapsed]
948 aws_eks_node_group.eks_node_group: Still creating... [02m30s elapsed]
949 aws_eks_node_group.eks_node_group: Still creating... [02m40s elapsed]
950 aws_eks_node_group.eks_node_group: Still creating... [02m50s elapsed]
951 aws_eks_node_group.eks_node_group: Still creating... [03m00s elapsed]
952 aws_eks_node_group.eks_node_group: Still creating... [03m10s elapsed]
953 aws_eks_node_group.eks_node_group: Still creating... [03m20s elapsed]
954 aws_eks_node_group.eks_node_group: Still creating... [03m30s elapsed]
955 aws_eks_node_group.eks_node_group: Still creating... [03m40s elapsed]
956 aws_eks_node_group.eks_node_group: Still creating... [03m50s elapsed]
957 aws_eks_node_group.eks_node_group: Still creating... [04m00s elapsed]
958 aws_eks_node_group.eks_node_group: Still creating... [04m10s elapsed]
959 aws_eks_node_group.eks_node_group: Still creating... [04m20s elapsed]
960 aws_eks_node_group.eks_node_group: Still creating... [04m30s elapsed]
961 aws_eks_node_group.eks_node_group: Still creating... [04m40s elapsed]
962 aws_eks_node_group.eks_node_group: Creation complete after 4m40s [id=mundose-cluster:mundose-node-group]
963
964 Apply complete! Resources: 38 added, 0 changed, 0 destroyed.
965
966 Outputs:
967
968 cluster_endpoint = "https://9CBEDE2C23853B3D1EA510D2B9EAA780.q7.us-east-1.eks.amazonaws.com"
969 cluster_name = "mundose-cluster"
  
```

Recursos AWS creados:

The screenshot shows the AWS Management Console for the 'mundose-cluster' EKS cluster. The cluster is in the 'Active' state and is using the '1.28' version of Kubernetes. The console displays the cluster's name, state, version, and other details. The cluster is located in the 'us-east-1' region. The console also shows the cluster's endpoint and name.

Nombre del clúster	Estado	Versión de Kubernetes	Periodo de soporte	Política de actualización	Creado	Proveedor
mundose-cluster	Activo	1.28	Mejorar ahora	Soporte extendido hasta el 25 de noviembre de 2023	Extendido	hace una hora EKS

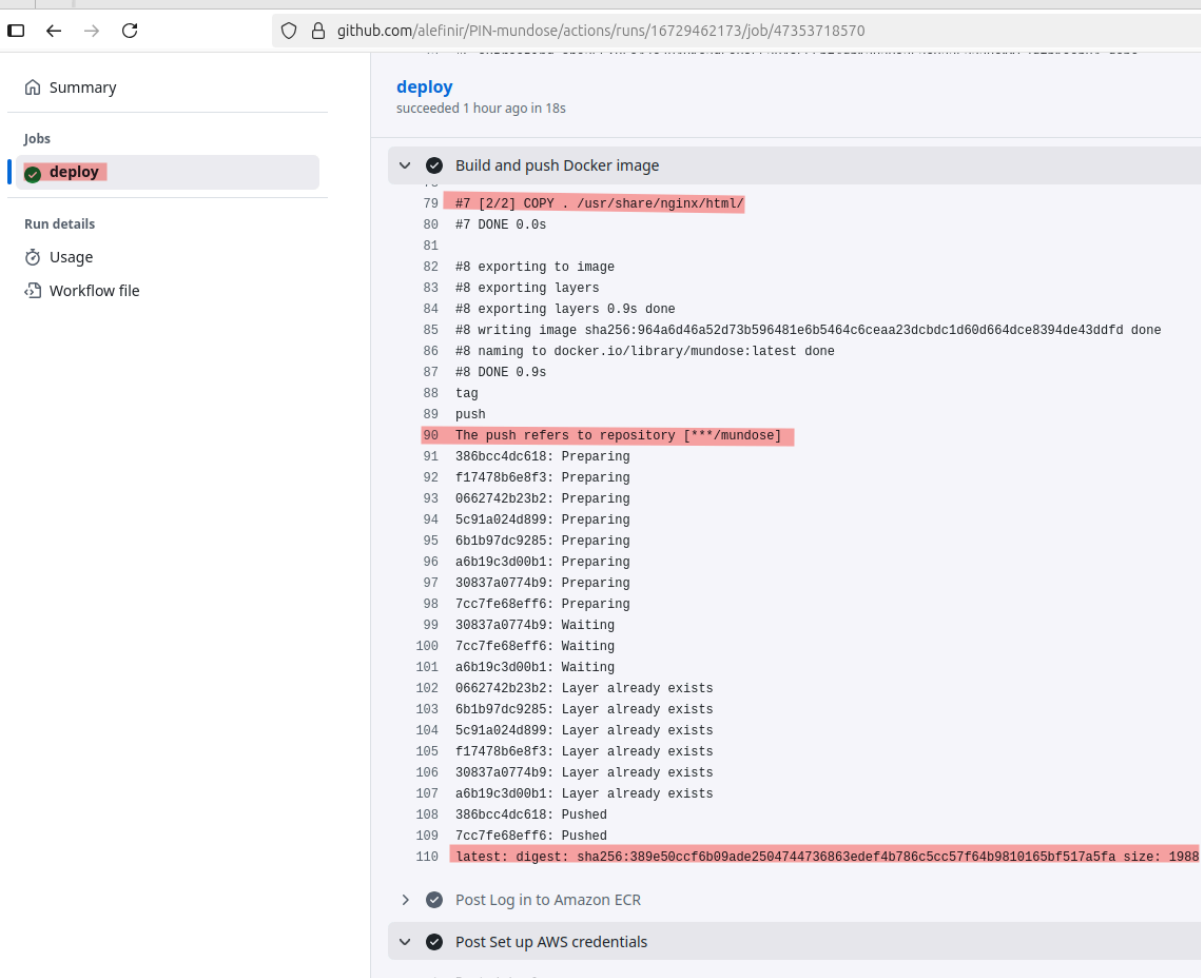
Link al repo y al video respectivo:

<https://github.com/alefinir/PIN-mundose/tree/main/terraform-backend>

<https://youtu.be/XmLvtXGzF9c>

Build Image and Upload it to ECR:

En este workflow se toman los archivos de la app (html, javascript y css) para crear una imagen custom de NGINX. Luego de creada la imagen esta es subida al repositorio de imágenes de AWS llamado ECR para su posterior uso por EKS.



The screenshot shows a GitHub Actions workflow run for the repository `alefinir/PIN-mundose`. The workflow is named `deploy` and has succeeded 1 hour ago in 18s. The main step is `Build and push Docker image`, which is expanded to show the following log output:

```

79 #7 [2/2] COPY . /usr/share/nginx/html
80 #7 DONE 0.0s
81
82 #8 exporting to image
83 #8 exporting layers
84 #8 exporting layers 0.9s done
85 #8 writing image sha256:964a6d46a52d73b596481e6b5464c6ceaa23dc9d6d64dce8394de43ddfd done
86 #8 naming to docker.io/library/mundose:latest done
87 #8 DONE 0.9s
88 tag
89 push
90 The push refers to repository [***/mundose]
91 386bcc4dc618: Preparing
92 f17478b6e8f3: Preparing
93 0662742b23b2: Preparing
94 5c91a024d899: Preparing
95 6b1b97dc9285: Preparing
96 a6b19c3d00b1: Preparing
97 30837a0774b9: Preparing
98 7cc7fe68eff6: Preparing
99 30837a0774b9: Waiting
100 7cc7fe68eff6: Waiting
101 a6b19c3d00b1: Waiting
102 0662742b23b2: Layer already exists
103 6b1b97dc9285: Layer already exists
104 5c91a024d899: Layer already exists
105 f17478b6e8f3: Layer already exists
106 30837a0774b9: Layer already exists
107 a6b19c3d00b1: Layer already exists
108 386bcc4dc618: Pushed
109 7cc7fe68eff6: Pushed
110 latest: digest: sha256:389e50ccf6b09ade2504744736863edef4b786c5cc57f64b9810165bf517a5fa size: 1988
  
```

Below the main step, there are two other steps: `Post Log in to Amazon ECR` and `Post Set up AWS credentials`, both of which are also completed successfully.



The screenshot shows the Amazon Elastic Container Registry (ECR) console. The left sidebar shows the navigation menu with options for `Private registry` and `Public registry`. The main content area displays a list of private repositories. The repository `mundose` is highlighted, showing its details:

Nombre del repositorio	URI	Creado en	Inmutabilidad de etiqueta	Tipo de cifrado
<code>mundose</code>	<code>061470250089.dkr.ecr.us-east-1.amazonaws.com/mundose</code>	02 de agosto de 2025, 21:50:06 (UTC-03)	Mutable	AES-256

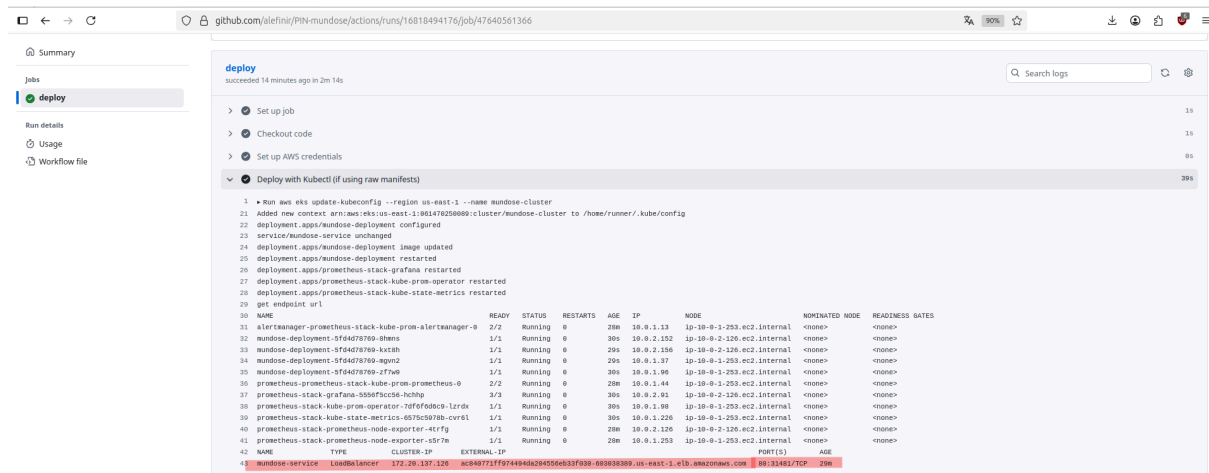
Link al repo y al video respectivo:

<https://github.com/alefinir/PIN-mundose/tree/main/app>

<https://youtu.be/hn8hdMCGITE>

Deploy Application to EKS:

En esta parte se hace un deploy de la imagen al cluster creado previamente, ahí se pueden ver los distintos pods que soportan la aplicación y el resultado final.



deploy
succeeded 14 minutes ago in 2m 14s

Jobs

- Summary
- Jobs
 - deploy
- Run details
- Usage
- Workflow file

1 > Run aws eks update-kubeconfig --region us-east-1 --name mundose-cluster

21 Added new context: arn:aws:eks:us-east-1:865478250899:cluster/mundose-cluster to /home/runner/.kube/config

22 deployment.apps/mundose-deployment configured

23 service/mundose-service unchanged

24 deployment.apps/mundose-deployment image updated

25 deployment.apps/mundose-deployment restarted

26 deployment.apps/prometheus-stack-grafana restarted

27 deployment.apps/prometheus-stack-kube-prom-operator restarted

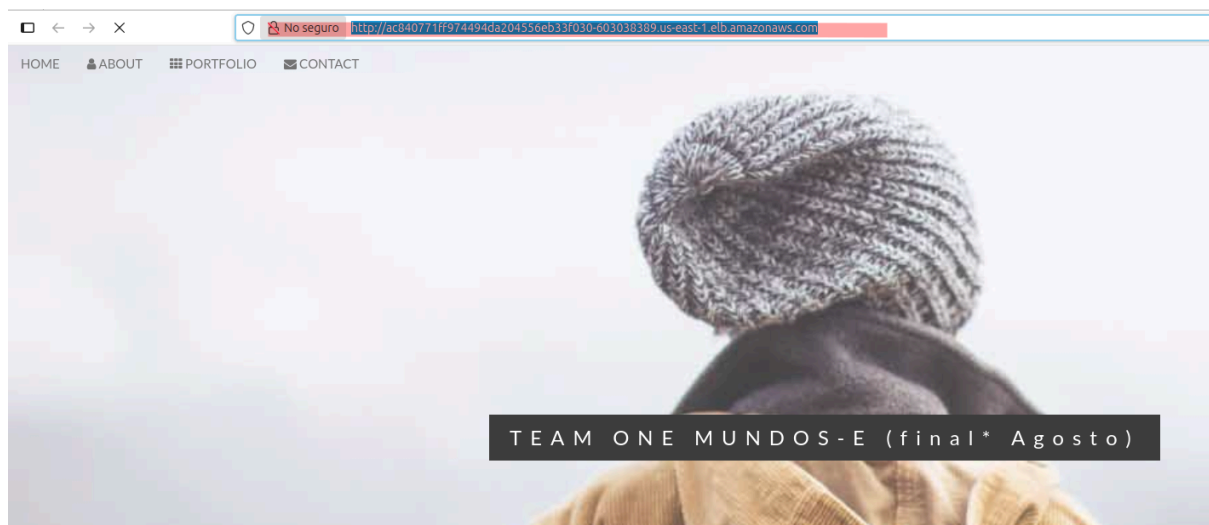
28 deployment.apps/prometheus-stack-kube-state-metrics restarted

29 get endpoint url

NAME	READY	STATUS	RESTARTS	AGE	IP	MODE	NOTREADY NODE	READINESS GATES
alermanager-prometheus-stack-kube-prom-alertmanager-0	2/2	Running	0	28m	10.0.1.13	ip-10-0-1-253.ec2.internal	<none>	<none>
mundose-deployment-5f64278789-bhms	1/1	Running	0	30s	10.0.2.152	ip-10-0-2-126.ec2.internal	<none>	<none>
mundose-deployment-5f64278789-kat8h	1/1	Running	0	29s	10.0.2.156	ip-10-0-2-126.ec2.internal	<none>	<none>
mundose-deployment-5f64278789-ajm2l	1/1	Running	0	29s	10.0.1.37	ip-10-0-1-253.ec2.internal	<none>	<none>
mundose-deployment-5f64278789-zf7wz	1/1	Running	0	30s	10.0.1.96	ip-10-0-1-253.ec2.internal	<none>	<none>
prometheus-prometheus-stack-kube-prom-prometheus-0	2/2	Running	0	28m	10.0.1.44	ip-10-0-1-253.ec2.internal	<none>	<none>
prometheus-stack-grafana-556f5c5c-hcmg	3/3	Running	0	30s	10.0.2.161	ip-10-0-2-126.ec2.internal	<none>	<none>
prometheus-stack-kube-prom-operator-7df8f6d9c9-lzdx	1/1	Running	0	30s	10.0.1.98	ip-10-0-1-253.ec2.internal	<none>	<none>
prometheus-stack-kube-state-metrics-6575c5978b-cvrl	1/1	Running	0	30s	10.0.1.226	ip-10-0-1-253.ec2.internal	<none>	<none>
prometheus-stack-kube-state-metrics-6575c5978b-cvrl	1/1	Running	0	28m	10.0.2.126	ip-10-0-2-126.ec2.internal	<none>	<none>
prometheus-stack-kube-state-metrics-6575c5978b-cvrl	1/1	Running	0	28m	10.0.1.253	ip-10-0-1-253.ec2.internal	<none>	<none>

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
mundose-service	LoadBalancer	172.20.137.126	ac840771ff974494da204556eb33f030-603030389.us-east-1.elb.amazonaws.com	80:31481/TCP	29s

URL de la Aplicación Deployada:



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Deploy del Monitoreo:

En el mismo deploy se puede observar la instalación de las herramientas de monitoreo (Prometheus + Grafana) en el mismo cluster EKS con Helm. Se corren un par de comandos más para obtener la URL de Grafana y el password de acceso correspondiente.

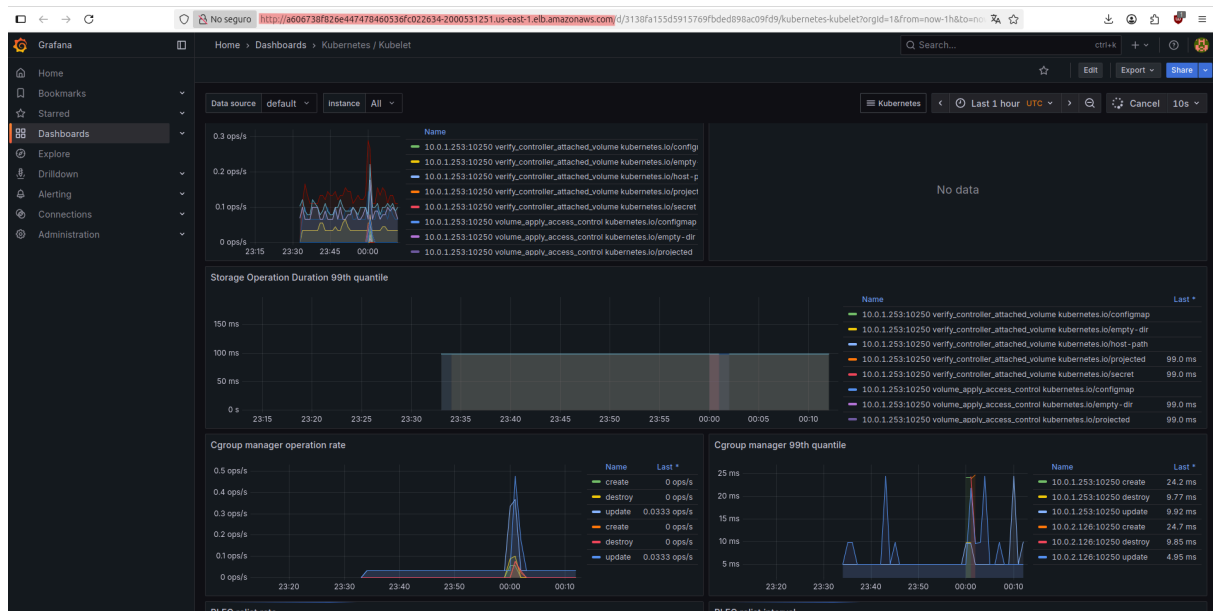
The screenshot shows a GitHub Actions workflow run for deploying Prometheus and Grafana to an EKS cluster. The workflow is titled "Deploy Prometheus and Grafana stack" and is located in the repository "alefinir/PIN-mundose".

The workflow steps include:

- Install Helm:** This step installs Helm on the EKS cluster. It uses the "linux-and64/README.md" file to get the latest version of Helm.
- Deploy Prometheus and Grafana stack:** This step deploys the Prometheus and Grafana stack to the EKS cluster. It uses the "prometheus-community/prometheus" chart to deploy Prometheus and the "grafana/grafana" chart to deploy Grafana.

The workflow output shows the status of the deployment. The Prometheus stack is deployed successfully, and the Grafana admin password is generated. The output also shows the URL of the Prometheus stack and the Grafana admin password.

URL del Monitoreo:



Link al repo y al video respectivo:

<https://github.com/alefinir/PIN-mundose/tree/main/eks>

<https://youtu.be/Fcr3tGBHLIE>

Instancia EC2 base de operaciones:

Esta instancia se creó por medio de Terraform para administrar el cluster y correr comandos manuales de acuerdo a las necesidades.

github.com/alefinir/PIN-mundose/actions/runs/16832587396/job/47683824174

Summary

Jobs

terraform_apply

Run details

Usage

Workflow file

terraform_apply

succeeded 50 minutes ago in 36s

Terraform apply

```

102     ]
103
104     + capacity_reservation_specification (known after apply)
105
106     + cpu_options (known after apply)
107
108     + ebs_block_device (known after apply)
109
110     + enclave_options (known after apply)
111
112     + ephemeral_block_device (known after apply)
113
114     + instance_market_options (known after apply)
115
116     + maintenance_options (known after apply)
117
118     + metadata_options (known after apply)
119
120     + network_interface (known after apply)
121
122     + private_dns_name_options (known after apply)
123
124     + root_block_device (known after apply)
125   }
126
127   Plan: 1 to add, 0 to change, 0 to destroy.
128
129   Changes to Outputs:
130     + public_ip      = (known after apply)
131   aws_instance.mundose_vm: Creating...
132   aws_instance.mundose_vm: Still creating... [00m10s elapsed]
133   aws_instance.mundose_vm: Creation complete after 13s [id=i-037a1a483f6d92105]
134
135   Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
136
137   Outputs:
138
139   cluster_endpoint = "https://5250A8A77EC92B45E12DD5F8449291DD.gr7.us-east-1.eks.amazonaws.com"
140   cluster_name     = "mundose-cluster"
141   public_ip        = "54.146.208.156"
  
```

> Complete job

aws

Buscar [Alt+S]

Estados Unidos (Norte de Virginia) ID de cuenta

EC2 > Instancias

EC2

Panel

Vista global de EC2

Eventos

Instancias

Tipos de instancia

Instancias (1/3) Información

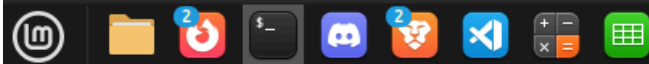
Conectar Estado de la instancia Acciones Lanzar instancia

Buscar instancia por atributo o etiqueta (case-sensitive) Todos los ...

	Name	ID de la instancia	Estado de la instancia	Tipo de instancia	Comprobación de salud	Estado de la instancia	Zona de disponibilidad	DNS de IPv4 pública	Dirección IP pública	IP privada
		i-07b750e9376081618	En ejecución	t3.medium	3/3 comprobador	Ver alarmas +	us-east-1b	-	54.234.184.184	-
		i-0d4c35f9f1df3ea13	En ejecución	t3.medium	3/3 comprobador	Ver alarmas +	us-east-1a	-	13.221.159.99	-
☒	mundose-vm	i-037a1a483f6d92105	En ejecución	t2.micro	2/2 comprobador	Ver alarmas +	us-east-1b	ec2-54-146-208-156.co...	54.146.208.156	-

Ejemplo de comandos de administración sobre cluster EKS desde la instancia EC2:

```
[root@ip-172-31-10-61 ec2-user]# kubectl version --client
kubectl get nodes
Client Version: v1.33.3
Kustomize Version: v5.6.0
NAME                                STATUS    ROLES    AGE    VERSION
ip-10-0-1-215.ec2.internal          Ready    <none>    55m    v1.28.15-eks-3abbec1
ip-10-0-2-188.ec2.internal          Ready    <none>    55m    v1.28.15-eks-3abbec1
[root@ip-172-31-10-61 ec2-user]# kubectl delete deployments --all --all-namespaces
deployment.apps "mundose-deployment" deleted
deployment.apps "prometheus-stack-grafana" deleted
deployment.apps "prometheus-stack-kube-prom-operator" deleted
deployment.apps "prometheus-stack-kube-state-metrics" deleted
deployment.apps "coredns" deleted
[root@ip-172-31-10-61 ec2-user]# kubectl delete services --all --all-namespaces
service "alertmanager-operated" deleted
service "kubernetes" deleted
service "mundose-service" deleted
service "prometheus-operated" deleted
service "prometheus-stack-grafana" deleted
service "prometheus-stack-kube-prom-alertmanager" deleted
service "prometheus-stack-kube-prom-operator" deleted
service "prometheus-stack-kube-prom-prometheus" deleted
service "prometheus-stack-kube-state-metrics" deleted
service "prometheus-stack-prometheus-node-exporter" deleted
service "eks-extension-metrics-api" deleted
service "kube-dns" deleted
service "prometheus-stack-kube-prom-coredns" deleted
service "prometheus-stack-kube-prom-kube-controller-manager" deleted
service "prometheus-stack-kube-prom-kube-etcd" deleted
service "prometheus-stack-kube-prom-kube-proxy" deleted
service "prometheus-stack-kube-prom-kube-scheduler" deleted
service "prometheus-stack-kube-prom-kubelet" deleted
[root@ip-172-31-10-61 ec2-user]# kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-10-0-1-215.ec2.internal          Ready    <none>    56m    v1.28.15-eks-3abbec1
ip-10-0-2-188.ec2.internal          Ready    <none>    56m    v1.28.15-eks-3abbec1
[root@ip-172-31-10-61 ec2-user]#
```



Terraform Destroy:

Por último se limpia toda la configuración y los recursos creados por medio de un terraform destroy, las URLs quedan inaccesibles.

```

1078 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 07m10s elapsed]
1079 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 07m20s elapsed]
1080 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 07m30s elapsed]
1081 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 07m40s elapsed]
1082 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 07m50s elapsed]
1083 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 08m00s elapsed]
1084 aws_eks_node_group.eks_node_group: Still destroying... [id=mundose-cluster:mundose-node-group, 08m10s elapsed]
1085 aws_eks_node_group.eks_node_group: Destruction complete after 08m15s
1086 aws_iam_role_policy_attachment.eks_registry_policy: Destroying... [id=eks-node-role/arn:aws:iam::aws:policy/AmazonEC2ContainerRegistryReadOnly]
1087 aws_iam_role_policy_attachment.eks_node_policy: Destroying... [id=eks-node-role/arn:aws:iam::aws:policy/AmazonEKSWorkerNodePolicy]
1088 aws_iam_role_policy_attachment.eks_cni_policy: Destroying... [id=eks-node-role/arn:aws:iam::aws:policy/AmazonEKS_CNI_Policy]
1089 aws_eks_cluster.eks_cluster: Destroying... [id=mundose-cluster]
1090 aws_iam_role_policy_attachment.eks_cni_policy: Destruction complete after 1s
1091 aws_iam_role_policy_attachment.eks_registry_policy: Destruction complete after 1s
1092 aws_iam_role_policy_attachment.eks_node_policy: Destruction complete after 1s
1093 aws_iam_role.eks_node_role: Destroying... [id=eks-node-role]
1094 aws_iam_role.eks_node_role: Destruction complete after 0s
1095 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 08m10s elapsed]
1096 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 08m20s elapsed]
1097 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 08m30s elapsed]
1098 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 08m40s elapsed]
1099 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 08m50s elapsed]
1100 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 09m00s elapsed]
1101 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 09m10s elapsed]
1102 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 09m20s elapsed]
1103 aws_eks_cluster.eks_cluster: Still destroying... [id=mundose-cluster, 09m30s elapsed]
1104 aws_eks_cluster.eks_cluster: Destruction complete after 9m35s
1105 aws_iam_role_policy_attachment.eks_cluster_policy: Destroying... [id=eks-cluster-role/arn:aws:iam::aws:policy/AmazonEKSClusterPolicy]
1106 aws_subnet.public_subnet_b: Destroying... [id=subnet-058ce3bc6f224941]
1107 aws_subnet.public_subnet_a: Destroying... [id=subnet-04bac75b2965213e0]
1108 aws_iam_role_policy_attachment.eks_cluster_policy: Destruction complete after 0s
1109 aws_iam_role.eks_cluster_role: Destroying... [id=eks-cluster-role]
1110 aws_iam_role.eks_cluster_role: Destruction complete after 0s
1111 aws_subnet.public_subnet_b: Destruction complete after 1s
1112 aws_subnet.public_subnet_a: Destruction complete after 1s
1113 aws_vpc.eks_vpc: Destroying... [id=vpc-071187e1d2745c1dd]
1114 aws_vpc.eks_vpc: Destruction complete after 0s
1115
1116 Destroy complete! Resources: 38 destroyed.
1117
> Complete job

```

URL de la APP

```

http://ac840771ff974494da204556eb33f030-603038389.us-east-1.elb.amazonaws.com

```

Uf. Tenemos problemas para encontrar ese sitio.

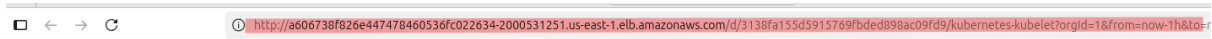
No podemos conectar al servidor en ac840771ff974494da204556eb33f030-603038389.us-east-1.elb.amazonaws.com.

Si escribió la dirección correcta, puede:

- Probar de nuevo más tarde
- Verificar la conexión a internet
- Comprobar que Firefox tiene permiso para acceder a la web (puede ser que esté conectado pero detrás de un firewall)

Reintentar

URL del Monitoreo



Uf. Tenemos problemas para encontrar ese sitio.

No podemos conectar al servidor en a606738f826e447478460536fc022634-2000531251.us-east-1.elb.amazonaws.com.

Si escribió la dirección correcta, puede:

- Probar de nuevo más tarde
- Verificar la conexión a internet
- Comprobar que Firefox tiene permiso para acceder a la web (puede ser que esté conectado pero detrás de un firewall)

Reintentar

Link al repo y al video respectivo:

<https://github.com/alefinir/PIN-mundose/tree/main/terraform-backend>

<https://youtu.be/tH2Wpylpytw>